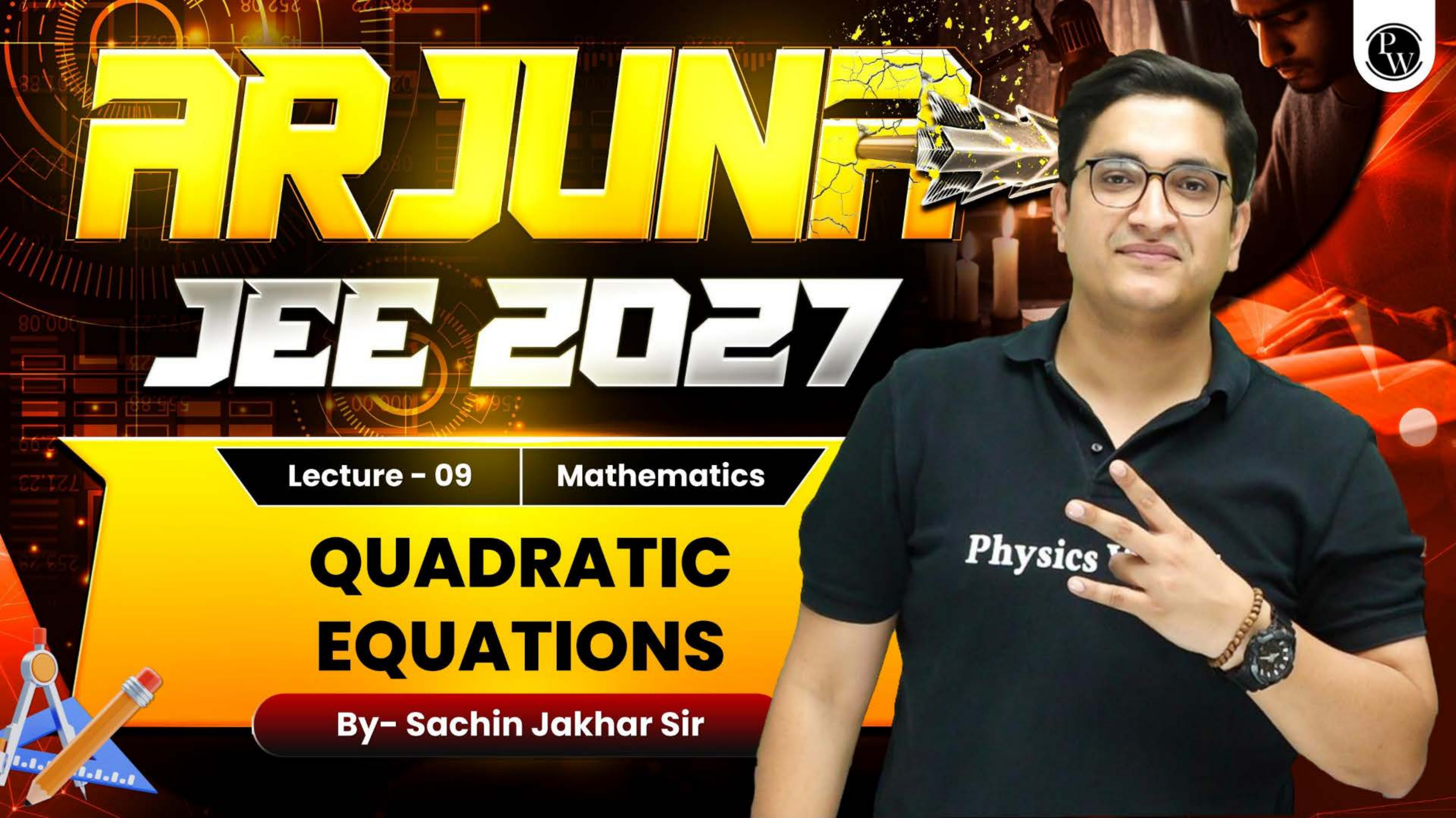
# ARJUNA JEE 2027

Lecture - 09 | Mathematics

# QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



# Topics *to be covered*

A Range of Quadratic

B Range of Rational Functions

C Question Practice

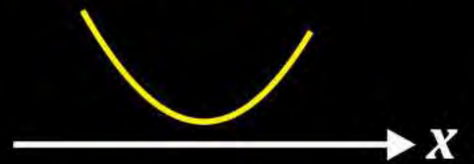
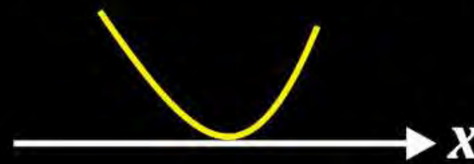
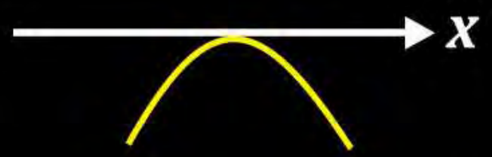
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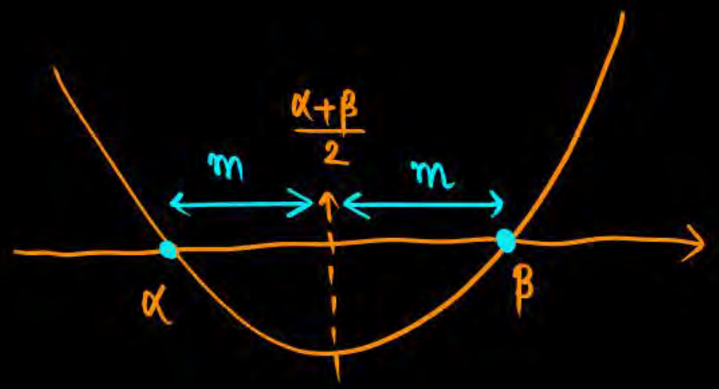


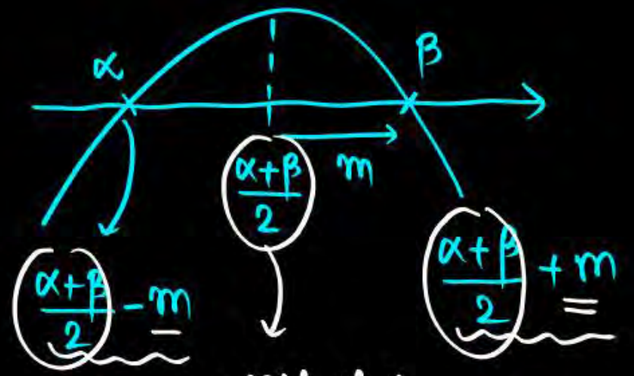
# LAST CLASS REVISION

1 → P → C
2 → S → D
3 → S, R → E
4 → P, Q → B

## # Answer the following Questions:

| Column-I | Column-II | Column-III |
|---|---|---|
| 1. $ax^2 + bx + c > 0 \; \forall x \in R$ | P.  | A. $a > 0, D \geq 0$ |
| 2. $ax^2 + bx + c < 0 \; \forall x \in R$ (for all) | Q.  | B. $a > 0, D \leq 0$ |
| 3. $ax^2 + bx + c \leq 0 \; \forall x \in R$ | R.  | C. $a > 0, D < 0$ |
| 4. $ax^2 + bx + c \geq 0 \; \forall x \in R$ | S.  | D. $a < 0, D < 0$ |
| | | E. $a < 0, D \leq 0$ |




$$\frac{-\text{coeff. of } x}{2\,\text{coeff. of } x^2}$$



# LAST CLASS REVISION

## #PO-SHEN-LOH Technique:

**Factorise following Quadratic:**

(i) $x^2 - 48x + 527 = 0$ → 31, 17

(ii) $x^2 - 78x + 1517 = 0$ → 37, 41

(iii) $2x^2 - 43x + 221 = 0$

$\left\{13, \frac{17}{2}\right\}$

Sum of Root $= \frac{43}{2}$

$\frac{43}{4} + \frac{9}{4} = \frac{52}{4} = 13$

Root $= \left(\frac{43}{4} + m\right), \left(\frac{43}{4} - m\right)$

$m = \frac{9}{4}$

$\frac{43}{4} - \frac{9}{4} = \frac{34}{4} = \frac{17}{2}$

$\left(\frac{43}{4} + m\right)\left(\frac{43}{4} - m\right) = \frac{221}{2}$

$\left(\frac{43}{4}\right)^2 - m^2 = \frac{221}{2}$

$\frac{43^2}{16} - \frac{221}{2} = m^2 = \frac{1849 - 1768}{16} = \frac{81}{16}$

$m = \pm\frac{9}{4}$

$\frac{\alpha+\beta}{2} = \frac{48}{2} = 24$

Root $= 24 + m, 24 - m$

$24 + 7, 24 - 7$ → 31, 17

Prod. $= 24^2 - m^2 = 527$

$576 - 527 = m^2$

$49 = m^2$

$\pm 7 = m$

# Question Practice

## Question [JEE Mains-2020]

54%

**Let $f(x)$ be a quadratic polynomial such that $f(-1)+f(2)=0$. If one of the roots of $f(x)=0$ is 3, then its other root lies in:**

- **A** $(-1, 0)$
- **B** $(-3, -1)$
- **C** $(0, 1)$
- **D** $(1, 3)$

$f(x) = ax^2+bx+c$

$x=-1$: $f(-1) = a-b+c$

$x=2$: $f(2) = 4a+2b+c$

$x=3$: $9a+3b+c=0$

$a-b+c + 4a+2b+c = 0$

$5a + b + 2c = 0$ — (1)

$9a + 3b + c = 0$ — (2)

$2\times$ (2): $18a + 6b + 2c = 0$

$5a + b + 2c = 0$

$13a + 5b = 0$

$\# \; b = -\frac{13}{5}a$

$5a - \frac{13}{5}a = -2c$

$\frac{6a}{5} = -c$

$-\frac{6a}{5} = c$

$$f(x) = ax^2 + bx + c$$

$$ax^2 - \frac{13}{5}ax - \frac{6a}{5} = 0$$

$$a\left\{x^2 - \frac{13}{5}x - \frac{6}{5}\right\} = 0$$

$$\frac{a}{5}\left\{5x^2 - 13x - 6\right\} = 0$$

$$\frac{a}{5}\left(5x^2 - 15x + 2x - 6\right) = 0$$

$$\frac{a}{5}(5x+2)(x-3) = 0$$

$x = 3$ ✓

$x = -\frac{2}{5}$

M-2

$x=3$ is a root

$x=\alpha$ is another root

Say :-

$$f(x) = k(x-3)(x-\alpha)$$

$$f(-1) + f(2) = 0$$

$$k(-1-3)(-1-\alpha) + k(2-3)(2-\alpha) = 0$$

$$k\{-4(-1-\alpha) + (-1)(2-\alpha)\} = 0$$

$$k\{4+4\alpha-2+\alpha\} = 0$$

$$k(2+5\alpha) = 0$$

$$\alpha = -2/5 \checkmark$$

## Acidic Question

Graph

**Comment on nature of roots of quadratic $f(x) = ax^2 + bx + c$, (where $a, b, c \in R$) for given conditions.**

**(i)** $a \cdot c < 0$ **(ii)** $(4a + 2b + c) \cdot (a - b + c) < 0$ **(iii)** $b^2 > (a + c)^2$

Sumit:

$D = b^2 - 4ac$ (-ve)

↓

+ve

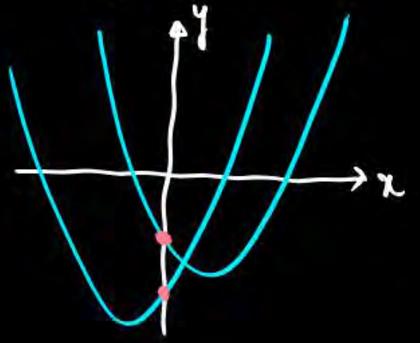
# Real & distinct.

Santosh: $ac < 0$

⇓

a & c are of opp. sign

$a > 0$ & $c < 0$




$a < 0$, $c > 0$



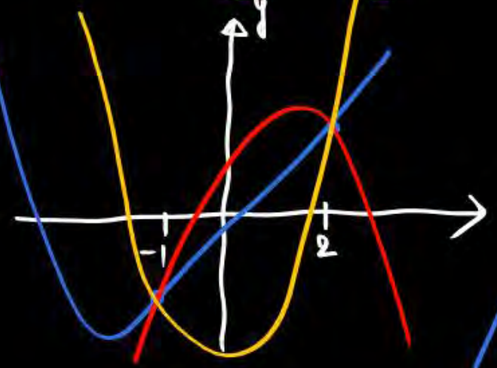
(ii) (value at $x = 2$) · (value at $x = -1$)

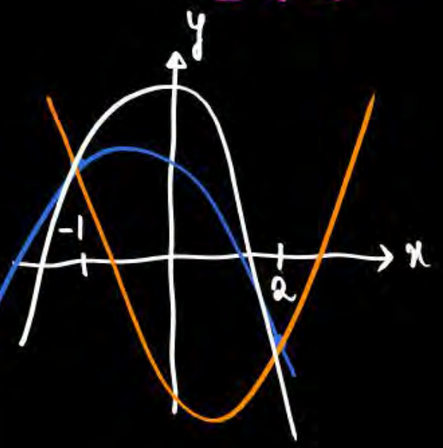
$f(2) \cdot f(-1) < 0$ → Real & dist.

→ 1 Root lies b/w -1 & 2.

⊕ ⊖

⊖ ⊕



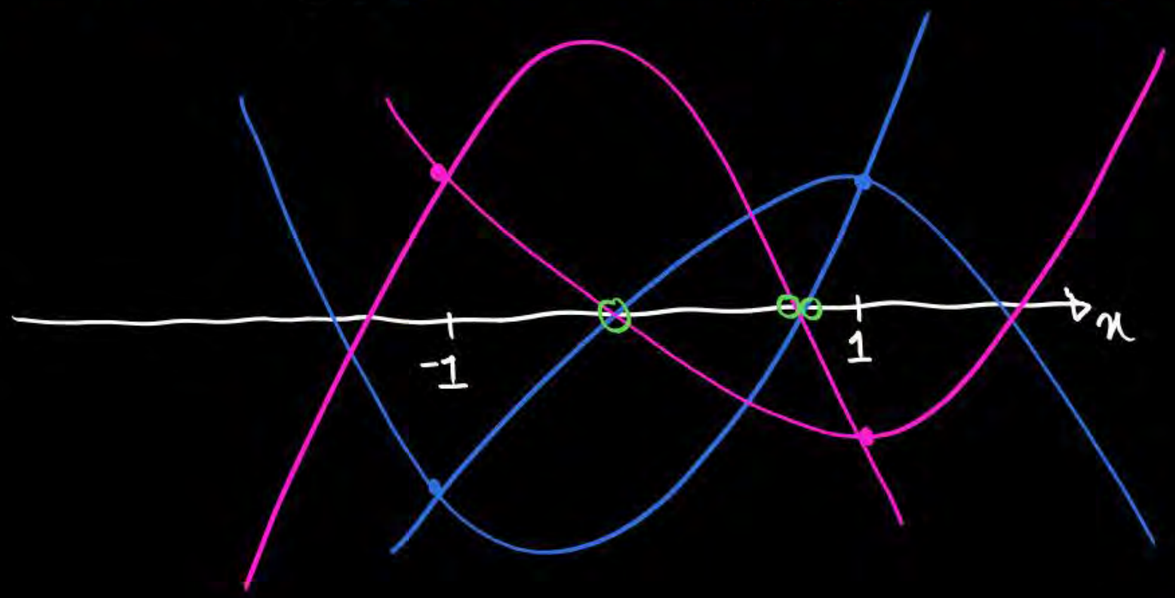



## Acidic Question

Graph

**Comment on nature of roots of quadratic $f(x) = ax^2 + bx + c$, (where $a, b, c \in R$) for given conditions.**

**(i)** $a \cdot c < 0$ **(ii)** $(4a + 2b + c) \cdot (a - b + c) < 0$ **(iii)** $b^2 > (a + c)^2$




$\Rightarrow b^2 - (a+c)^2 > 0$

$\Rightarrow (b+a+c)(b-a-c) > 0$

$(a+b+c)(a-b+c) < 0$

\# $f(1) \cdot f(-1) < 0$

\+ −

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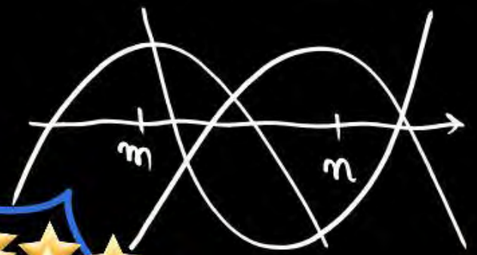
\# R & D

\# 1 Root lies b/w -1 & 1.

If $f(x)$ = Quadratic

and

$$f(m) \cdot f(n) < 0$$




# R+D Roots

# 1 root lies b/w $m$ & $n$

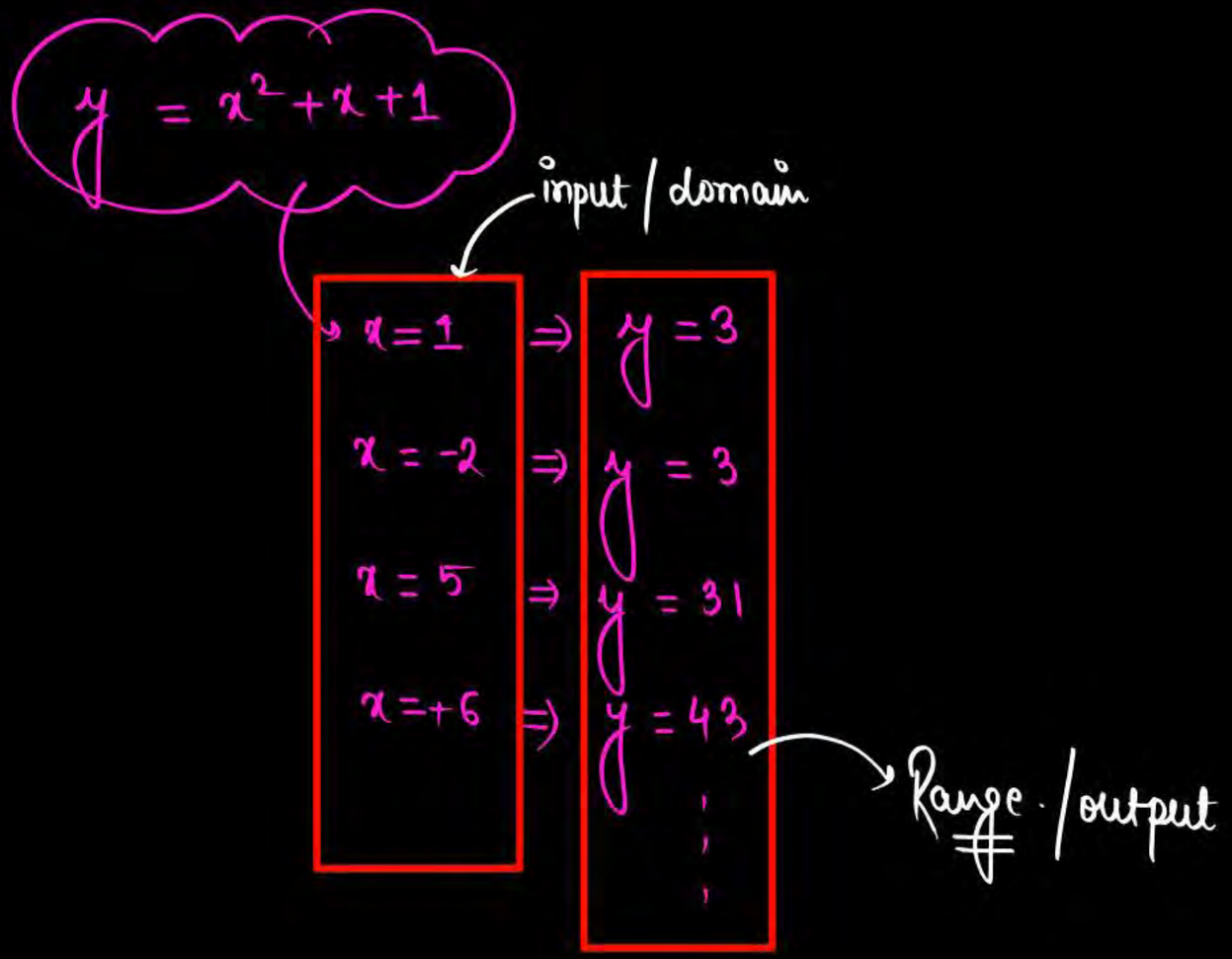
## Challenger Question [IIT-JEE]

If $a < b < c$ and $a, b, c \in R$ then comment on nature of roots of following:

(i) $f(x) = (x - a)(x - b) - 1$

(ii) $f(x) = (x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a)$




$y = x^2 + x + 1$
input / domain
$x = 1 \Rightarrow y = 3$
$x = -2 \Rightarrow y = 3$
$x = 5 \Rightarrow y = 31$
$x = +6 \Rightarrow y = 43$
Range / output

## IMPORTANT NOTE

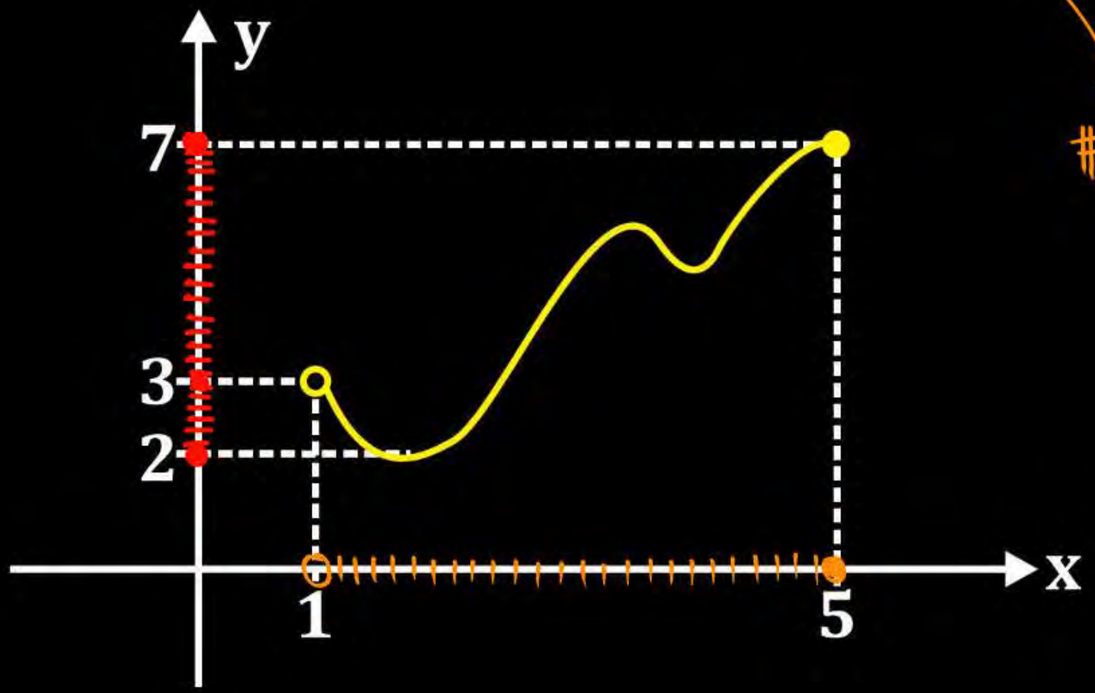
**Graphically**

**Domain (x)** ≡ portion of x-axis covered

**Range (y)** ≡ portion of y-axis covered

# FIND DOMAIN & RANGE FOR FOLLOWING GRAPHS

(i) $y = f(x)$



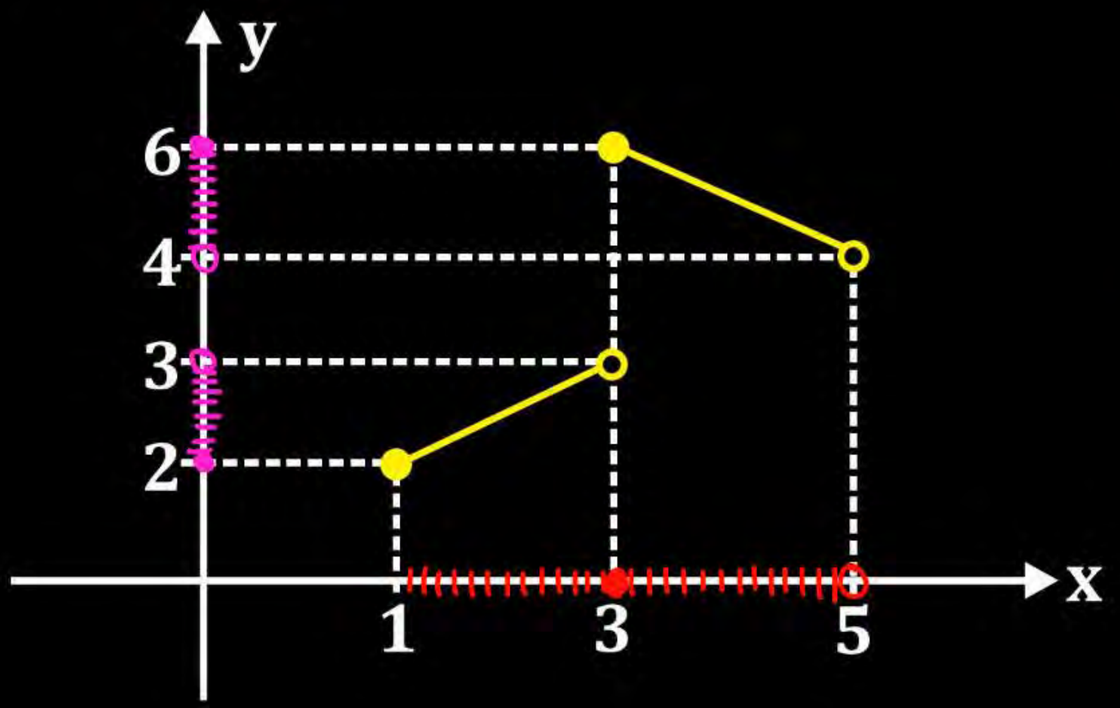

# domain(x) $\in (1, 5]$

Range → $y \in [2,7] - \{3\}$ ✗

Range → $y \in [2,7]$ ✓

# FIND DOMAIN & RANGE FOR FOLLOWING GRAPHS

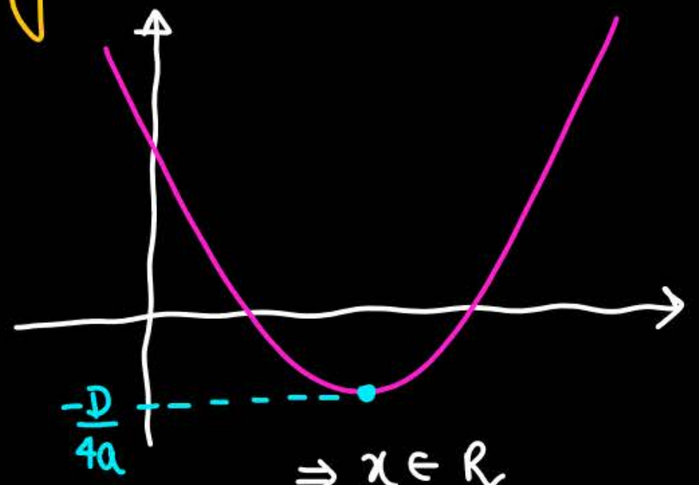
(ii) $y = f(x)$




# $x \in [1, 5)$

# $y \in [2, 3) \cup (4, 6]$

$y = ax^2 + bx + c$, $a > 0$

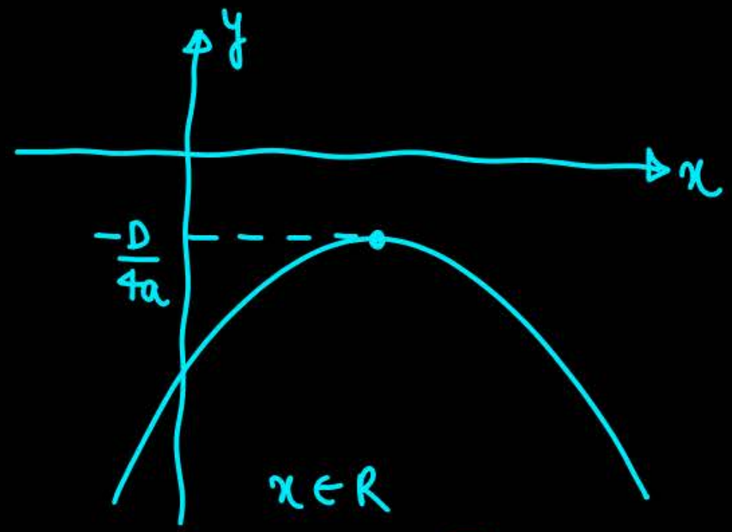



$\Rightarrow x \in R$

$\text{Range} \in \left[\frac{-D}{4a}, \infty\right)$ ✓

min. max.

$a < 0$




$x \in R$

$y \in \left(-\infty, \frac{-D}{4a}\right]$

Min. Max.

Ques:-

① Min. value of $y = 2x^2 - 3x + 1$ is $-\frac{1}{8}$.

$a > 0 \Rightarrow y \in \left[\frac{-D}{4a}, \infty\right)$

$y \in \left[\frac{-1}{4\times 2}, \infty\right)$

$y \in \left[\frac{-1}{8}, \infty\right)$

min.

② If $\alpha, \beta$ are roots of $x^2 - (a-1)x + (3a+5) = 0$ then min. value of $\alpha^2 + \beta^2 =$ ____

SBS

$\alpha + \beta = a - 1$

$\alpha\beta = 3a + 5$.

$\alpha^2 + \beta^2 + 2\alpha\beta = (a-1)^2$

$\alpha^2 + \beta^2 = (a-1)^2 - 2(3a+5)$

$= a^2 - 2a + 1 - 6a - 10$

$= a^2 - 8a - 9$.

min. value of $(a^2 - 8a - 9) = -25$.

# Quad.

Range $\in [-25, \infty)$



# RANGE OF QUADRATIC

Quadratic: $y = ax^2 + bx + c$

(i) If $a > 0 \Rightarrow$ $\text{Range}(y) \in \left[-\frac{D}{4a}, \infty\right)$

(ii) If $a < 0 \Rightarrow$ $\text{Range}(y) \in \left(-\infty, -\frac{D}{4a}\right]$

## IMPORTANT NOTE

(i) Above formulas are only valid if $x \in R$.

(ii) In case of restricted domain, use graphical method to calculate Range.

## Question

**Find range of following functions:**

(i) $f(x) = 3x^2 - 2x - 7$

(ii) $f(x) = 3x^2 - 2x - 7, x \in (0, 5]$ → Restricted domain.

(iii) $f(x) = 3x^2 - 2x - 7, x \in [-6, -1]$

$V \equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$

$\left(\frac{1}{3}, \frac{-4\times 22}{4\times 3}\right) \equiv \left(\frac{1}{3}, \frac{-22}{3}\right)$

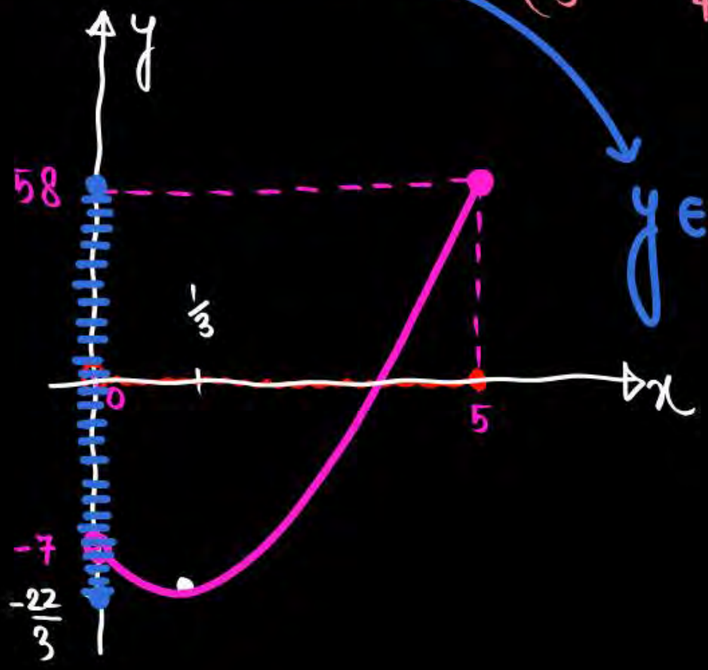
$a > 0$

(i) Range $\in \left[\frac{-22}{3}, \infty\right)$

$f(5) = 3\times 25 - 10 - 7$

$= 75 - 10 - 7$

$= 58.$




$y \in \left[\frac{-22}{3}, 58\right]$

## Question

**Find range of following functions:**

(i) $f(x) = 3x^2 - 2x - 7$

(ii) $f(x) = 3x^2 - 2x - 7, x \in (0, 5]$ → Restricted domain.

(iii) $f(x) = 3x^2 - 2x - 7, x \in [-6, -1]$

$V \equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$

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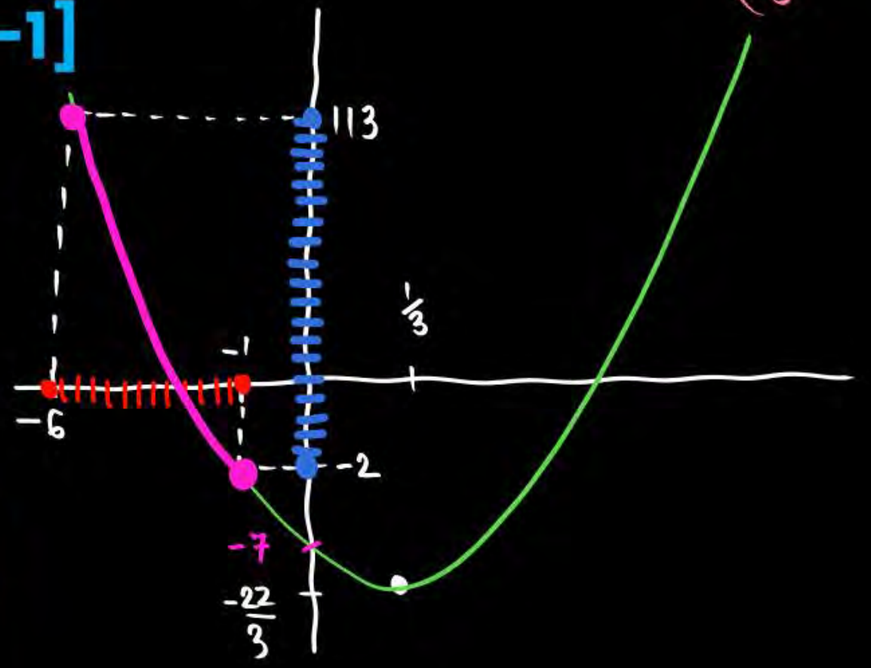
$a > 0$

(i) Range $\in \left[\frac{-22}{3}, \infty\right)$

$f(-6) = 3\times36 + 12 - 7$

$f(-1) = 3 + 2 - 7$

$= -2$




Range $\in [-2, 113]$

## Question

**Find range for following functions :**

(i) $f(x) = -x^2 + 4x - 6$

(ii) $f(x) = -x^2 + 4x - 6, x \in [-3, 1)$

(iii) $f(x) = -x^2 + 4x - 6, x \in (0, 4]$

Ans. : (i) $y \in (-\infty, -2]$, (ii) $y \in [-27, -3)$, (iii) $y \in [-6, -2]$

# RANGE OF RATIONAL FUNCTION

**Format-01 :** $\frac{\text{Linear}}{\text{Linear}}$ or $\frac{\text{Const.}}{\text{Linear}}$

$y = \frac{ax+b}{cx+d}$ ✓

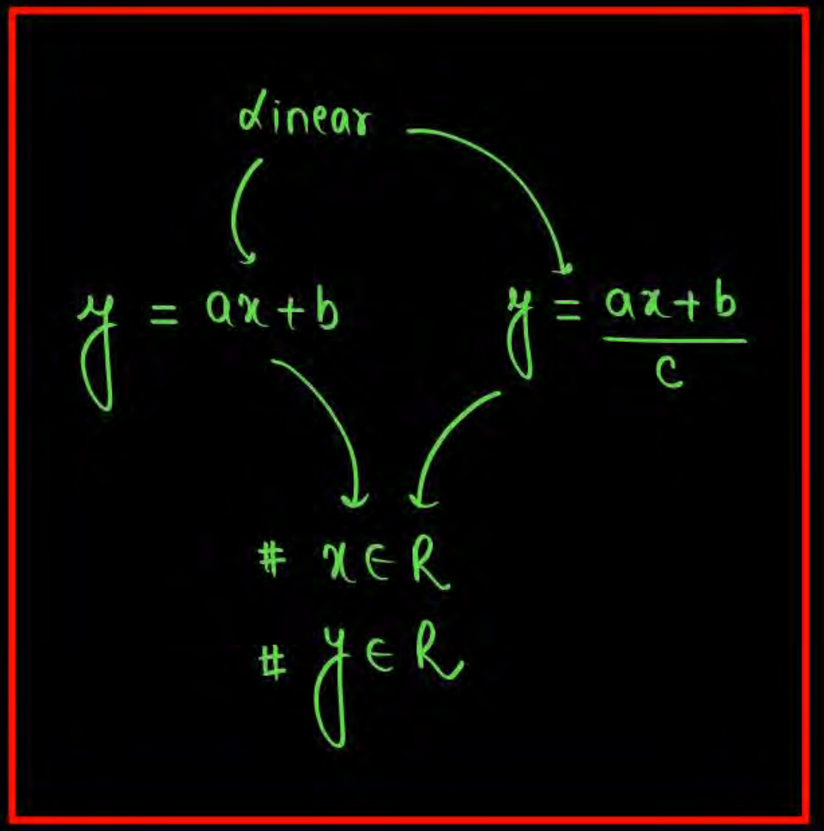
Domain$(x) \in R - \left\{\frac{-d}{c}\right\}$

Range$(y) \in R - \left\{\frac{a}{c}\right\}$

$y = \frac{b}{cx+d}$ ✓

$x \in R - \left\{\frac{-d}{c}\right\}$

$y \in R - \{0\}$



$y = \frac{ax+b}{cx+d}$

$y \in R - \left\{\frac{a}{c}\right\}$

$cyx + dy = ax + b$

$dy - b = ax - cyx$

$dy - b = x(a - cy)$

$\frac{dy-b}{a-cy} = x$

$a - cy \neq 0$

$\frac{a}{c} \neq y$

**Question**

## Find range for following functions :

(i) $f(x) = \dfrac{3x+2}{x+4}$ → $x \in R - \{-4\}$, $y \in R - \{3\}$

(ii) $f(x) = \dfrac{x-1}{2-7x}$ → $x \in R - \{\frac{2}{7}\}$, $y \in R - \{\frac{-1}{7}\}$

(iii) $f(x) = \dfrac{4}{3x+8}$ → $x \in R - \{\frac{-8}{3}\}$, $y \in R - \{0\}$

(iv) $f(x) = \dfrac{4x+5}{4}$ → $x \in R$, $y \in R$

(v) $f(x) = \dfrac{2x-5}{15-6x}$

(v) Impost.

$f(x) = \dfrac{2x-5}{15-6x}$

# $x \in R - \{\frac{5}{2}\}$ ✓

$y \in R - \{-\frac{1}{3}\}$ ✗

$y = \dfrac{2x-5}{-3(2x-5)}$

$y = \dfrac{-1}{3}$ $\quad y \in \{-\frac{1}{3}\}$ ✓

# DO IT BY YOURSELF (DIBY)

# DIBY-36

Consider the graph $f(x) = ax^2 + bx + c$ and comment on sign of following parts:

(i) $(abc)$ is _____.

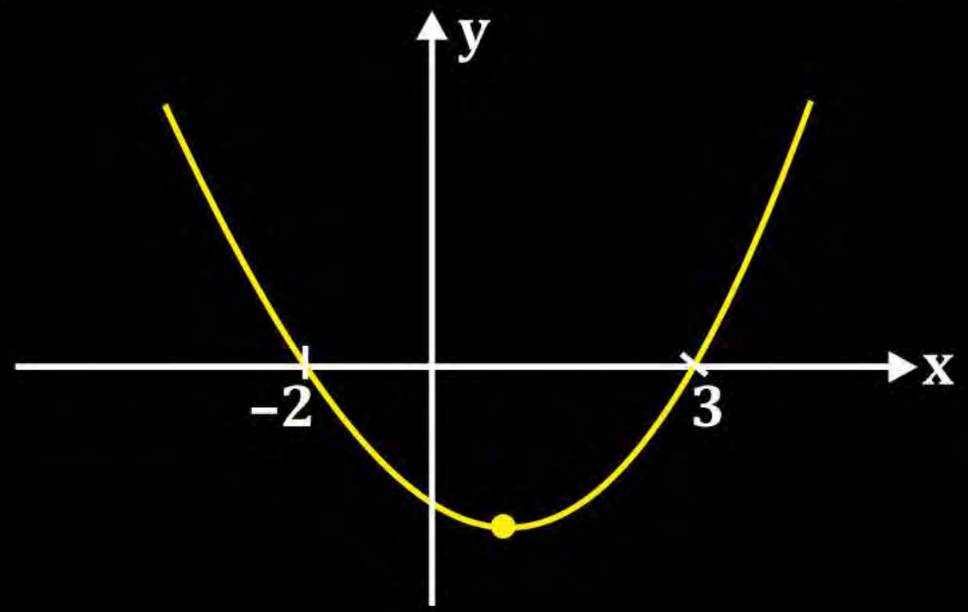
(ii) $(4a + 2b + c)$ is _____.

(iii) for all $x > 3, f(x)$ is _____.

(iv) For all $-2 < x < 3, f(x)$ is _____.

(v) $D$ is _____.

(vi) $(a - b + c)\cdot(9a - 3b + c)$ is _____.




**Ans. :**

(i) +ve, (ii) –ve, (iii) +ve, (iv) –ve, (v) +ve, (vi) –ve

# GRAPH OF QUADRATIC

## # DIBY-37(Challenger)

Draw the graph of quadratic $(y = ax^2 + bc + c)$ in each part with given conditions:

(i) Roots are 3 & 7 and crosses y-axis at (0, −2)

(ii) Imaginary roots with vertex at $\left(-\frac{3}{4}, \frac{1}{2}\right)$ and $a - b + c = 2$.

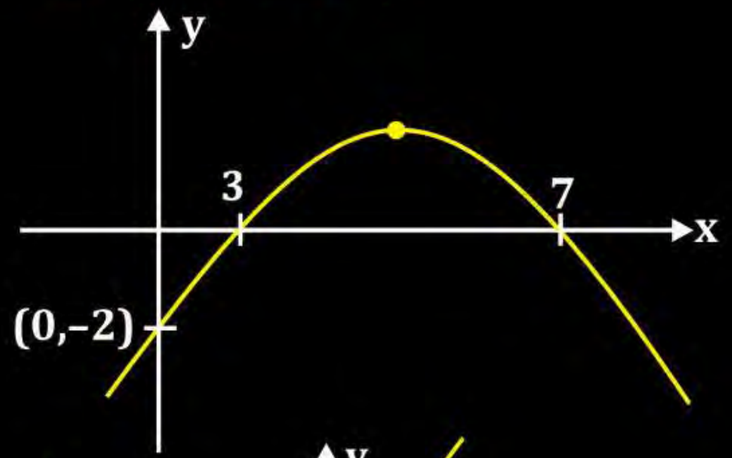
(iii) Touches x-axis at $x = -3$ and value at $x = 2$ is −8.

Also find value of $a, b$ & $c$ in each case.

# GRAPH OF QUADRATIC

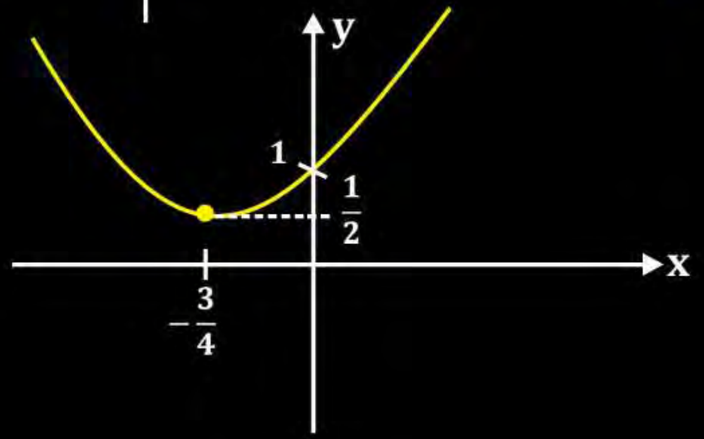
## # DIBY-37(Answer)

(i)



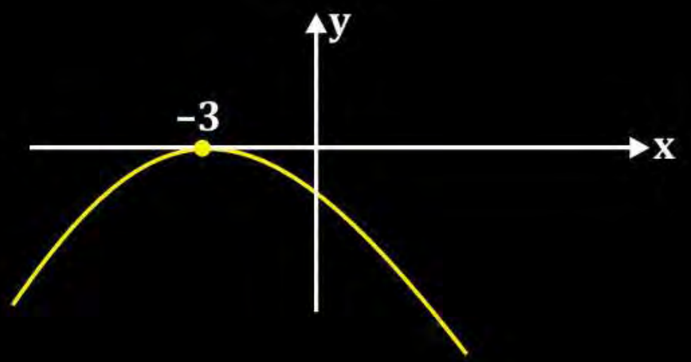

$a = -\frac{2}{21}, b = \frac{20}{21}, c = -2$

(ii)




$a = 24, b = 36, c = 14$

(iii)




$a = \frac{-8}{25}, b = \frac{-48}{25}, c = \frac{-72}{25}$

## # DIBY-38

If the equation $ax^2 + bc + c = 0, (a > 0)$ has two real roots $\alpha$ and $\beta$ such that $\alpha < -2$ and $\beta > 2$, then which of the following statements is/are true?

(A) $a - |b| + c < 0$

(B) $c < 0,\ b^2 - 4ac > 0$

(C) $4a - 2|b| + c < 0$

(D) $9a - 3|b| + c < 0$

[Ans. : A, B, C]

## # DIBY-39

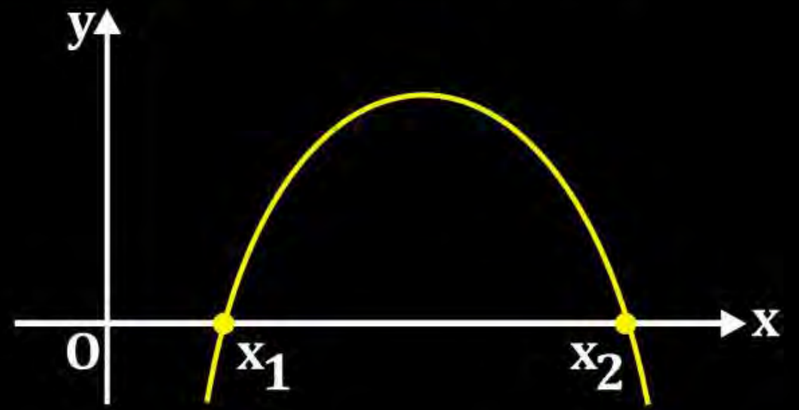
The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then

(A) $a < 0$

(B) $b^2 < 4ax$

(C) $c > 0$

(D) $a$ and $b$ are of opposite sign




[Ans. : A, D]

# RANGE OF QUADRATIC

**# DIBY-40**

**Find the range of $f(x)$ :**

**(i)** $f(x) = 2x^2 - 3x + 2$

**(ii)** $f(x) = 2x^2 - 3x + 2, x \in [0, 2]$

**[Ans. : (i) $\left[\frac{7}{8}, \infty\right)$, (ii) $\left[\frac{7}{8}, 4\right]$]**

**# DIBY-41** (done!)

**Find range of following functions:**

**(i)** $f(x) = 3x^2 - 2x - 7$

**(ii)** $f(x) = 3x^2 - 2x - 7, x \in (0, 5]$

**(iii)** $f(x) = 3x^2 - 2x - 7, x \in [-6, -1]$

**[Ans. : (i) $\left[-\frac{22}{3}, \infty\right)$, (ii) $[-22/3, 58]$, (iii) $[-2, 113]$]**



# Homework

# Re-attempt all Class Examples

# DPP & DIBY

# Module: (3 sittings of 1 HR each)

Exercise-02(Prabal): 1 to 10,13,16,18 to 25.

Exercise-03(Setu): 1 to 30 (except-13,16,17,21,25,26,29)

*It is never about end result*

*It's always about Journey*

# Thank You

#futureIITians